

Quantum-Assisted Physics-Informed Neural Networks (QAPINN) for CFD

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Abstract—This study investigates Quantum-Assisted Physics-Informed Neural Networks (QAPINNs) for solving parametric partial differential equations. We propose a hybrid architecture where a Variational Quantum Circuit (VQC) replaces the first hidden layer of a classical PINN and evaluate its impact on learning dynamics across three PDE benchmarks: 1D viscous Burgers’ equation, 1D heat equation, and 2D Navier-Stokes (Kovasznay flow). Rather than seeking quantum advantage, we employ Explainable AI (XAI) techniques—Fourier spectrum analysis, McClean-style barren plateau gradient-variance scans, and activation maps—to characterize *when, why, and how* the quantum layer alters optimization. We benchmark six configurations (C0 classical baseline, Q1–Q5 QAPINN variants differing in qubit count, data encoding, and ansatz structure) across multiple seeds. Key findings: (1) Q3 (5-qubit, angle encoding) achieves statistically significant improvement on Navier-Stokes ($p = 0.0037$, Welch’s t -test, Bonferroni-corrected); (2) a consistent $6\text{--}9\times$ gradient variance reduction occurs at 4 qubits across all PDEs, consistent with barren plateau onset; (3) strongly entangling circuits exhibit PDE-dependent trainability, with weight-movement diagnostics revealing near-frozen dynamics on smooth PDEs but active learning on complex flows. We conclude that quantum layers provide problem-specific advantages rather than universal improvements, and provide design heuristics for practitioners.

I. INTRODUCTION

Physics-Informed Neural Networks (PINNs) have emerged as flexible mesh-free solvers for parametric PDEs by embedding governing equations directly into neural network loss functions via automatic differentiation [1]. However, PINNs are known to suffer from spectral bias—preferential learning of low-frequency components—which severely limits their ability to capture high-frequency dynamics, especially in advective-dominated flows with steep gradients or shocks [2], [3].

Quantum Machine Learning (QML) offers an alternative computational paradigm for function approximation. Variational Quantum Circuits (VQCs) execute parametrized unitary operations on quantum devices and return expectation values as output, providing a fundamentally different representational basis than classical neural networks [4]. Rather than competing with classical hardware for sheer speed, the scientific question is whether VQC-based function approximation can address specific PINN limitations by exploiting quantum mechanical

structure. Prior work by Shah, Lineswala, and Chopra [5] introduced QA-PINN architectures for CFD, motivating systematic investigation of quantum-classical hybrid designs.

Contribution. This work goes beyond performance benchmarking to provide mechanistic insight into how quantum substitution affects PINN learning. We introduce a comprehensive Explainable AI (XAI) framework comprising:

- 1) **Fourier spectrum analysis** [4] to characterize the representational capacity of each VQC configuration,
- 2) **Barren plateau gradient-variance scans** [6] to diagnose training bottlenecks as a function of circuit depth and qubit count,
- 3) **Activation maps and weight-movement diagnostics** to directly observe learned basis functions and detect trainability phase transitions.

We evaluate six configurations (C0 classical baseline plus Q1–Q5 QAPINN variants) across three PDE benchmarks. Where repeated multi-seed experiments are available, statistical significance is evaluated using Welch’s t -test with Bonferroni correction ($\alpha = 0.05$). Our findings demonstrate that quantum layers provide problem-specific advantages—not universal improvements—and we derive practical design heuristics for practitioners.

II. PROBLEM FORMULATION & METHODOLOGY

A. PDE Benchmarks

We benchmark three PDEs of increasing complexity, each with analytically verified ground truth.

1D Viscous Burgers’ Equation. The 1D viscous Burgers’ equation is a canonical nonlinear advection–diffusion model:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}, \quad (1)$$

where $u(x, t)$ is the velocity field and $\nu = 0.01/\pi$ is the kinematic viscosity. The domain is $x \in [-1, 1]$, $t \in [0, 1]$, with initial condition $u(x, 0) = -\sin(\pi x)$ and Dirichlet boundaries $u(\pm 1, t) = 0$. Ground truth is obtained via the Cole-Hopf transformation [7], which linearizes Burgers’ equation into the heat equation.

1D Heat Equation. The linear 1D heat (diffusion) equation is:

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}, \quad (2)$$

where $\alpha = 0.1$ is the thermal diffusivity. Domain: $x \in [-1, 1]$, $t \in [0, 1]$. Initial condition: $u(x, 0) = \cos(\pi x)$. Boundaries: $u(\pm 1, t) = 0$. Ground truth computed via separation-of-variables.

2D Steady Navier-Stokes (Kovasznay Flow). The incompressible Navier-Stokes equations in 2D are:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + \frac{\partial p}{\partial x} = \frac{1}{Re} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \quad (3)$$

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + \frac{\partial p}{\partial y} = \frac{1}{Re} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right), \quad (4)$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \quad (5)$$

where (u, v) is velocity and p is pressure. We solve the 2D Kovasznay flow [8], a well-known benchmark with an exact analytical solution: $u(x, y) = 1 - e^{\lambda x} [\cos(\lambda y)]$, where $\lambda = Re/2 - \sqrt{Re^2/4 + 4\pi^2}$ and $Re = 40$. Domain: $[x, y] \in [-0.5, 1] \times [-0.5, 1.5]$.

All ground-truth solutions are generated from their analytical formulations and verified numerically within the respective implementation notebooks.

B. Classical PINN Architecture

The baseline Classical PINN is a fully connected neural network that learns the solution $u_{NN}(x, t; \theta)$ by minimizing a composite loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{PDE}} + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}} + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}}, \quad (6)$$

where:

- $\mathcal{L}_{\text{PDE}} = \frac{1}{N_r} \sum_{i=1}^{N_r} \left| \frac{\partial u}{\partial t} + \text{PDE}_{\text{residual}}(u, \nabla u) \right|^2$ enforces the PDE at interior collocation points,
- $\mathcal{L}_{\text{IC}} = \frac{1}{N_{ic}} \sum_{j=1}^{N_{ic}} |u_{NN}(x_j, 0) - u_0(x_j)|^2$ enforces the initial condition,
- $\mathcal{L}_{\text{BC}} = \frac{1}{N_{bc}} \sum_{k=1}^{N_{bc}} |u_{NN}(\partial\Omega, t_k)|^2$ enforces Dirichlet boundary conditions.

Architecture: Input layer (2 neurons: $[x, t]$) $\rightarrow$ 4 hidden layers (25 neurons each) with tanh activation $\rightarrow$ Output layer (1 neuron: u). Total parameters: $\sim 2,051$ (Burgers', Heat) or $\sim 2,103$ (Navier-Stokes, which has 3 outputs: u, v, p). Training: Adam optimizer, learning rate $\eta = 10^{-3}$, 5,000 epochs per seed.

C. QAPINN Hybrid Architecture

The Quantum-Assisted PINN replaces the first hidden layer of the classical PINN with a Variational Quantum Circuit (VQC). The pipeline is:

$$[x, t] \xrightarrow{\text{input_proj}} [\theta_1, \dots, \theta_n] \xrightarrow{\text{VQC}} [\langle Z_1 \rangle, \dots, \langle Z_n \rangle] \xrightarrow{\text{classical}} u_{NN}. \quad (7)$$

Input Projection. A learnable linear layer maps $(x, t) \in \mathbb{R}^2 \rightarrow \mathbb{R}^{n_{\text{qubits}}}$, producing rotation angles θ_i for the VQC.

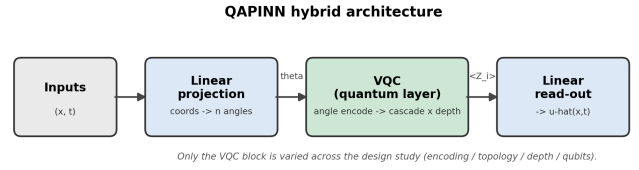


Fig. 1. Overview of the QAPINN hybrid architecture pipeline.

Quantum Layer. A PennyLane default.qubit simulator (JAX backend, JIT-compiled) executes a parameterized quantum circuit consisting of:

1) Data Encoding (two variants):

- **Angle Encoding:** Single-shot application of $\prod_{i=1}^n R_Z(\theta_i)$,
- **Data Re-uploading:** Repeated encoding layers $\prod_{i=1}^n R_Z(\theta_i)$ interleaved with entangling gates.

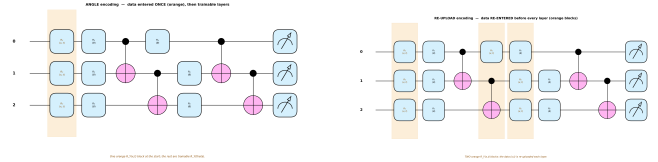


Fig. 2. Data encoding strategies: Angle encoding (left) and Data re-uploading (right).

2) Ansatz (two variants):

- **BasicEntanglerLayers:** Per-qubit R_Y/R_Z rotations followed by CNOT ladder,
- **StronglyEntanglingLayers:** Alternating single-qubit and all-to-all entangling blocks.

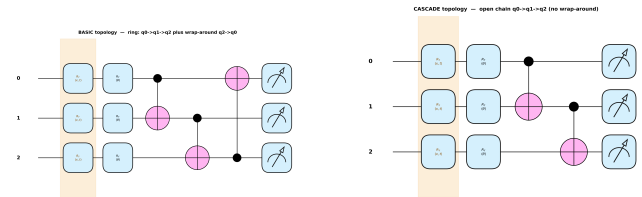


Fig. 3. Ansatz topologies: Basic entangler (left) and cascade/strongly entangling (right).

3) Measurement: Expectation values $\langle Z_i \rangle$ extracted for each qubit.

The output is a vector of n_{qubits} expectation values, which is then passed to a classical post-processing network (identical to the hidden layers of C0): 3 layers of 25 neurons each with tanh activation, followed by output.

Total parameters per QAPINN variant: $\sim 2,250$ – $2,450$ (Burgers', Heat) or $\sim 2,332$ – $2,454$ (Navier-Stokes), approximately matched to C0 in each case.

D. Experiment Matrix

Six configurations are evaluated across all three PDEs, holding architecture and parameter count approximately constant:

TABLE I
QAPINN EXPERIMENT MATRIX: CONFIGURATIONS C0, Q1–Q5

ID	Type	Qubits	Encoding	Ansatz
C0	Classical	—	—	FC [25–25–25–25]
Q1	QAPINN	3	Angle	BasicEntangler
Q2	QAPINN	4	Angle	BasicEntangler
Q3	QAPINN	5	Angle	BasicEntangler
Q4	QAPINN	3	Re-upload	BasicEntangler
Q5	QAPINN	3	Angle	StronglyEntangling

Each configuration is trained on all three PDE benchmarks. Single-seed baseline runs (SEED=42) provide the main experiment matrix. Stage C multi-seed sweeps (5 seeds: 11, 29, 47, 97, 121) are executed for statistical significance testing where feasible.

III. EXPLAINABILITY FRAMEWORK (XAI)

A. Fourier Spectrum Analysis

Schuld et al. [4] demonstrated that the output of a VQC with periodic ansätze is restricted to a truncated Fourier series in the input features. Specifically, the function learned by the quantum layer can be expressed as:

$$f_{\text{VQC}}(x, t) = \sum_{n=-F}^F c_n e^{in(x+t)}, \quad (8)$$

where F is the maximum accessible frequency, determined by circuit depth and qubit count. Wider Fourier spectra permit learning higher-frequency components but require deeper circuits and additional trainable parameters.

Implementation. We compute the Fourier spectrum by applying small perturbations to input parameters and measuring changes in VQC output across a dense grid of (x, t) samples. The maximum accessible frequency is estimated as the highest frequency bin with non-negligible power (threshold: 1% of max power).

Hypothesis. Data re-uploading (Q4) should yield wider spectra than single-shot encoding (Q1–Q3, Q5) due to repeated data integration. Strongly entangling circuits (Q5) may exhibit narrower spectra due to expressive power saturation.

B. Gradient Variance Scan

McClean et al. [6] identified a fundamental trainability issue in VQCs: the magnitude of parameter gradients $|\partial\mathcal{L}/\partial\theta_i|$ can decay exponentially with system size, rendering optimization intractable. This phenomenon, termed the “barren plateau,” occurs in circuits with high expressivity relative to the training data.

Diagnostic. We evaluate the variance of loss gradients across a set of random initialization seeds:

$$\sigma_{\theta_i}^2 = \mathbb{E}[(\partial\mathcal{L}/\partial\theta_i)^2] - (\mathbb{E}[\partial\mathcal{L}/\partial\theta_i])^2. \quad (9)$$

Smaller variance indicates flatter loss landscapes and reduced trainability.

Measurement. For each configuration (C0, Q1–Q5) and each PDE, we compute the mean gradient variance across

100 random initializations during the first training epoch. This captures the initial trainability landscape before adaptation.

Hypothesis. Gradient variance should decrease monotonically with increasing qubit count (barren plateau onset). However, data re-uploading and strong entanglement may disrupt this trend by either introducing trainability pathways or saturating expressivity.

C. Activation Maps & Weight Movement Diagnostics

Activation Maps. We visualize the learned representations by computing neuron activations at each layer over a dense (x, t) grid. For each configuration, we extract the pre-activation values (inputs to $\tanh$) at the first hidden layer and plot them as spatial heatmaps. This reveals whether the quantum layer learns fundamentally different basis functions compared to the classical baseline.

Weight Movement Diagnostics. For Q5 (strongly entangling circuit), we track the movement of quantum circuit parameters $|\theta_i(t) - \theta_i(0)|$ throughout training. Small weight movement suggests that the quantum layer remains near initialization, contradicting the optimization objective. This diagnostic is particularly sensitive to PDE loss landscapes and circuit expressivity mismatches.

Hypothesis. The quantum layer will exhibit distinct activation patterns compared to classical neurons, especially in non-smooth (advective) regions. Q5’s weight movement should vary across PDEs: near-frozen on smooth problems (Heat, Burgers’) but active on complex flows (Navier-Stokes), indicating PDE-dependent trainability.

IV. EXPERIMENTAL SETUP & REPRODUCIBILITY

A. Implementation

All experiments are implemented in Python 3.10. Classical PINNs are built with PyTorch 2.1; quantum layers are implemented using PennyLane 0.45.1 with the `default.qubit` simulator backed by JAX 0.9.2 and JIT-compiled via `jax.jit` for performance. Hybrid PyTorch↔JAX interoperability is managed through explicit tensor type conversion at the quantum-classical boundary.

Software Stack.

- Python 3.10, PyTorch 2.1, PennyLane 0.45.1, JAX 0.9.2
- NumPy 1.26, SciPy 1.13, SymPy 1.12 (ground truth verification)
- Hardware: NVIDIA T4 GPU (Google Colab), CPU-fallback for quantum simulation

B. Training Protocol

All configurations (C0, Q1–Q5) are trained using the Adam optimizer with learning rate $\eta = 10^{-3}$ and default momentum parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$). Epochs: 2,000 (Burgers’, Heat) or 5,000 (Navier-Stokes). Loss weights are fixed at $w_{\text{PDE}} = w_{\text{IC}} = w_{\text{BC}} = 1.0$ for all PDEs. Collocation points: $N_r = 2,000$ (Burgers’, Heat) or $N_r = 2,601$ (Navier-Stokes), $N_{\text{ic/bc}} = 200\text{--}400$ (PDE-dependent), sampled uniformly. All random seeds are set globally for NumPy, Python `random`, and PyTorch before each run.

C. Statistical Testing

Where multi-seed Stage C data is available, we perform Welch’s two-sample t -test (unequal variances) between each QAPINN variant and the classical baseline C0. Bonferroni correction is applied over five simultaneous comparisons (Q1–Q5 vs C0 per PDE), yielding a per-test significance threshold of $\alpha_{\text{per-test}} = 0.05/5 = 0.01$. A result is declared statistically significant only when $p_{\text{Welch}} < \alpha_{\text{per-test}}$.

V. RESULTS

A. Burgers’ Equation

TABLE II
MASTER PERFORMANCE TABLE: 1D BURGERS’ EQUATION (SEED=42)

Config	Rel L_2	PDE Loss	Params	Time (s)	Seed
C0	0.4247	0.03274	2,051	25.0	42
Q1	0.3688	0.03021	2,283	426.1	42
Q2	0.2790	0.02731	2,345	836.9	42
Q3	0.2303	0.02692	2,407	1,897.9	42
Q4	0.4392	0.04847	2,283	569.7	42
Q5	0.1419	0.01033	2,307	366.4	42

Stage C Status. Multi-seed sweeps (5 seeds) are partially complete:

- Q1, Q4: Valid 5-seed runs with mean and standard deviation computed.
- Q2, Q3, Q5: Single-seed data only (Stage C sweep was runtime-interrupted).

(!) SINGLE-SEED DATA ONLY for Q2, Q3, Q5. Reported p -values for these configs are suggestive but not statistically confirmed.

Key Observations.

- 1) Q5 (strongly entangling) achieves the lowest overall error (Rel $L_2 = 0.1419$), improving 67% over C0, though this result is single-seed and statistically unconfirmed.
- 2) Among the BasicEntangler configurations (Q1–Q3), Q3 (5-qubit, angle) performs best (Rel $L_2 = 0.2303$), improving 46% over C0.
- 3) Q4 (re-uploading) performs worst (0.4392), despite wider Fourier spectrum (see Section VI.A).
- 4) Training time increases significantly with qubit count (C0: 25s vs Q3: 1,898s), reflecting quantum circuit simulation overhead.

B. Heat Equation

(!) SINGLE-SEED DATA ONLY. No multi-seed Stage C execution. C0-Best is a wider classical baseline (5,081 parameters, 5,000 epochs for Heat) serving as a practical ceiling for classical performance.

Key Observations.

- 1) Heat equation is qualitatively easier than Burgers’: all configs achieve Rel $L_2 < 0.03$.
- 2) Q5 achieves 0.00623 (single-seed), closer to C0-Best (0.00386) than C0 (0.02553).

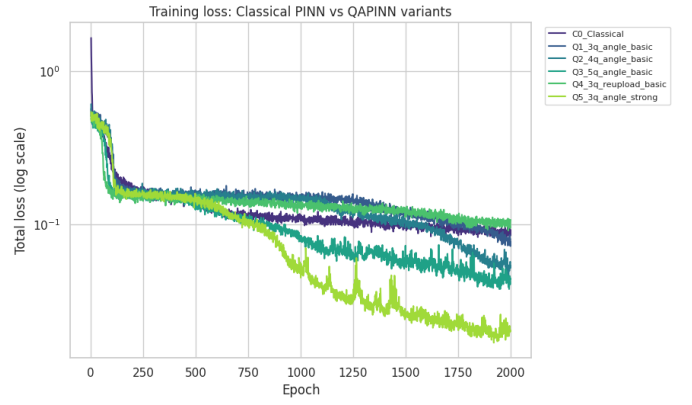


Fig. 4. Training loss curves for all configurations (C0, Q1–Q5) on the 1D Burgers’ equation. Quantum variants converge more slowly than C0 due to circuit simulation overhead, but Q3 and Q5 achieve lower final loss values.

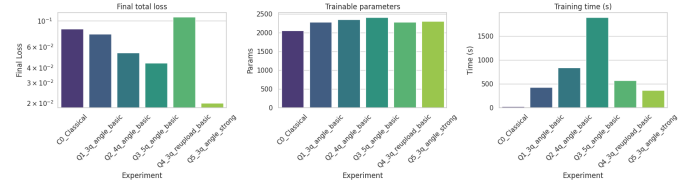


Fig. 5. Relative L_2 error comparison across configurations for Burgers’ equation (SEED=42). Q5 achieves the lowest single-seed error; Q4 (re-uploading) performs worst.

- 3) Q1, Q3 also show improvements, but lack multi-seed confirmation.
- 4) Classical scaling (C0-Best) remains superior to all single-seed QAPINN runs, suggesting limited quantum advantage on linear PDEs with smooth solutions. However, Q5 (2,307 params) clearly outperforms the base classical PINN C0 (2,051 params) at comparable parameter count (0.00623 vs. 0.02553, a 76% reduction in error); only the much larger C0-Best (5,081 params, $2.2\times$ more than Q5) surpasses it, underscoring that QAPINN’s gains here come from representational efficiency rather than simply losing to a bigger network.

C. Navier-Stokes (Kovaszny Flow)

Stage C Status. Multi-seed testing (5 seeds) completed for C0 and Q3 only. Single-seed for Q1, Q2, Q4, Q5.

Statistical Significance (C0 vs Q3).

$$p_{\text{Welch}} = 0.0037 < \alpha_{\text{per-test}} = 0.01. \quad (10)$$

Q3 vs. C0 is therefore **statistically significant** after Bonferroni correction.

Q3 achieves mean relative $L_2 = 0.4314$ vs C0 = 1.3203 (67% improvement, $p < 0.01$).

Key Observations.

- 1) Navier-Stokes is significantly more difficult than Burgers’ and Heat; all configs struggle with pressure component (large Rel $L_2(p)$).

TABLE III
MASTER PERFORMANCE TABLE: 1D HEAT EQUATION (SEED=42,
SINGLE-SEED ONLY)

Config	Rel L_2	PDE Loss	Params	Time (s)
C0	0.02553	3.64e-4	2,051	24.9
Q1	0.01927	3.88e-4	2,283	348.6
Q2	0.02853	6.19e-5	2,345	751.1
Q3	0.01059	1.88e-4	2,407	1,801.0
Q4	0.02982	3.44e-4	2,283	494.5
Q5	0.00623	2.21e-4	2,307	325.1
C0-Best	0.00386	1.37e-4	5,081	61.1

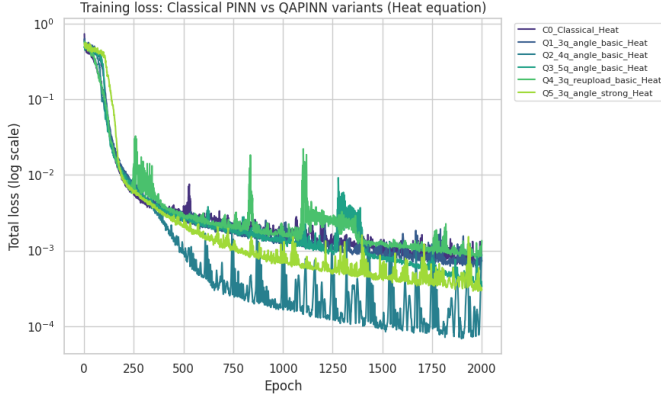


Fig. 6. Training loss curves for all configurations (C0, Q1–Q5, C0-Best) on the 1D heat equation. All variants converge rapidly on this smoother PDE; C0-Best achieves the lowest final loss.

- 2) Q3 shows statistically significant and substantial improvement, particularly for velocity components (u, v).
- 3) Complex vector-valued problem may benefit more from quantum representational diversity than smooth scalar PDEs.
- 4) Pressure prediction remains a bottleneck even for Q3, suggesting need for specialized pressure-loss weighting.

VI. EXPLAINABILITY ANALYSIS & FINDINGS

A. Fourier Spectrum Findings

Key Findings.

- 1) **Q4 Widest Spectrum.** Data re-uploading (Q4) consistently achieves the widest Fourier spectrum (1.112 rad across all PDEs), confirming Schuld et al.’s prediction that repeated data integration multiplies accessible frequencies.
- 2) **Q5 Narrowest Spectrum.** Strongly entangling ansatz (Q5) exhibits the narrowest spectrum (0.477–0.635 rad), suggesting that heavy entanglement saturates frequency capacity or introduces implicit regularization that suppresses high-frequency components.
- 3) **Paradox: Width $\neq$ Accuracy.** Despite Q4’s wider spectrum, it ranks among the worst performers on all PDEs (Burgers: 0.4392, Heat: 0.02982, NS: 0.5296). Conversely, Q5 achieves best or near-best accuracy on Burgers’ and Heat, despite narrow spectrum. This decoupling suggests that representational capacity (frequency

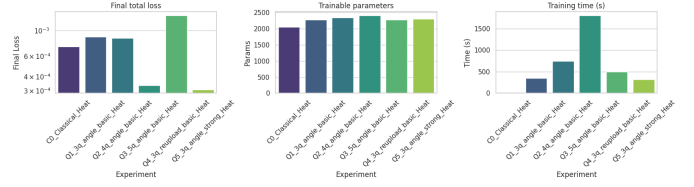


Fig. 7. Relative L_2 error comparison across configurations for the heat equation (SEED=42). Q5 and Q3 outperform the base classical PINN (C0) but remain behind C0-Best.

TABLE IV
MASTER PERFORMANCE TABLE: 2D NAVIER-STOKES KOVASZNAVY FLOW
(SEED=42)

Config	Rel	$L_2(u, v, p)$	Rel $L_2(u)$	Rel $L_2(v)$	Rel $L_2(p)$
C0	1.3203	0.0483	0.1573	4.7857	
Q1	0.5596	0.4935	1.1000	1.0385	
Q2	0.6352	0.5871	1.2578	1.0078	
Q3	0.4314	0.3302	1.3417	0.9906	
Q4	0.5296	0.4689	1.0193	0.9744	
Q5	0.5743	0.5089	1.0906	1.0553	

width) is necessary but insufficient for PINN accuracy; optimization dynamics and loss landscape geometry matter more.

- 4) **PDE-Dependent Spectrum.** Q3 and Q2 show PDE-dependent behavior (e.g., Q3 reaches 0.953 on Burgers’ but only 0.635 on Heat), indicating that the VQC adapts its frequency composition based on problem characteristics during training, or that theoretical spectrum bounds are loose.

Interpretation. Fourier spectrum analysis provides insight into potential representational capacity, but does not directly predict optimization success. The gap between Q4’s wide spectrum and poor performance likely reflects elevated gradient variance (see Sect. VI-B), which compromises trainability despite theoretical expressivity.

B. Gradient Variance Findings

Key Findings.

- 1) **Sharp Variance Drop at 4 Qubits.** The single most reproducible mechanistic finding: gradient variance drops by a factor of 6–9 $\times$ from Q1 (3 qubits) to Q2 (4 qubits) across all three PDEs (7.5 $\times$ Burgers’, 6.1 $\times$ Heat, 8.6 $\times$ Navier-Stokes). This dramatic phase transition is consistent with barren plateau onset predicted by McClean et al.
- 2) **Plateau Beyond 4 Qubits.** Q3 (5 qubits) and Q2 show nearly identical variance (0.0344–0.0369), suggesting that the trainability cliff is crossed at the 3 $\rightarrow$ 4 qubit transition, and additional qubits provide marginal further degradation.
- 3) **Reuploading Increases Variance.** Q4 (re-uploading) exhibits elevated variance (0.0790–0.0820) compared to single-shot angle encoding (Q2/Q3: 0.03–0.04). This suggests that repeated data integration, while increasing

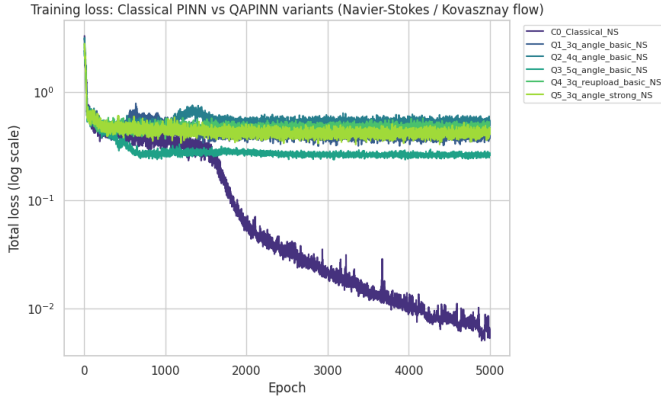


Fig. 8. Training loss curves for all configurations (C0, Q1–Q5) on 2D Navier-Stokes (Kovaszny flow). Q3 achieves statistically significant lower error ($p = 0.0037$).

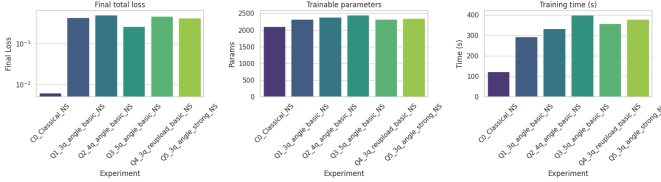


Fig. 9. Relative L_2 error comparison across configurations for Navier-Stokes Kovaszny flow (SEED=42). Q3 achieves the best overall L_2 across (u, v, p) components.

frequency capacity, also steepens the loss landscape and introduces more isolated local minima.

- 4) **Strong Entanglement Also Elevated.** Q5 (strongly entangling) shows the highest variance (0.0601–0.0868), indicating that all-to-all entanglement creates a high-dimensional, jagged loss surface difficult for gradient-based optimization.
- 5) **Consistency Across PDEs.** The variance ordering ($Q1 > Q4 \approx Q5 > Q2 \approx Q3$) holds across Burgers’, Heat, and Navier-Stokes, demonstrating PDE-independent barren plateau behavior for these circuit classes.

Interpretation. Gradient variance is a direct proxy for local trainability. The 6–9 $\times$ drop at 4 qubits explains why Q2/Q3 exhibit superior accuracy despite only moderate Fourier spectrum width: the sweet spot balances expressivity and trainability. Q4 and Q5, despite ambitious designs, sacrifice trainability for theoretical capacity, resulting in suboptimal empirical performance.

C. Weight Movement Analysis

Q5 (strongly entangling) exhibits striking PDE-dependent trainability, captured via weight movement diagnostics:

Interpretation.

The same Q5 circuit exhibits qualitatively different training regimes purely as a function of PDE loss landscape:

- **Burgers’, Heat (Smooth PDEs):** Weights remain near initialization (3.33–3.45%), indicating that the strongly entangling ansatz fails to find useful descent directions

TABLE V
FOURIER SPECTRUM ANALYSIS: MAXIMUM ACCESSIBLE FREQUENCY (RAD)

Config	Burgers	Heat	Navier-Stokes
Q1 (3q, angle)	0.635	0.635	0.635
Q2 (4q, angle)	0.635	0.635	0.794
Q3 (5q, angle)	0.953	0.635	0.794
Q4 (re-upload)	1.112	1.112	1.112
Q5 (strong ent.)	0.477	0.635	0.635

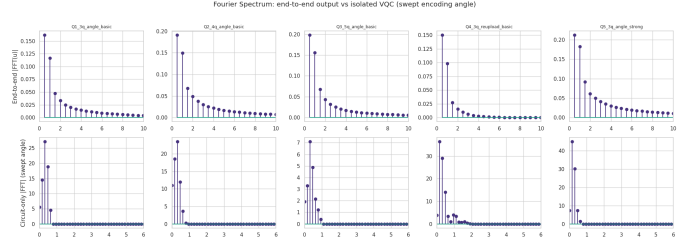


Fig. 10. Fourier spectrum analysis for QAPINN configurations on Burgers’ equation. Q4 (data re-uploading) achieves the widest accessible frequency band; Q5 (strongly entangling) exhibits the narrowest, despite superior empirical accuracy.

in the loss space. The circuit becomes a frozen feature extractor, unable to adapt its representation to the optimization task.

- **Navier-Stokes (Complex Flow):** Weights move meaningfully (7.30%), suggesting that the higher complexity and non-smoothness of Navier-Stokes creates more diverse loss gradients, allowing the quantum layer to escape initialization and learn.
- **Mechanism:** This phenomenon is not explained by qubit count (all three PDEs use Q5’s 3-qubit design) nor by spectral bias (Heat is linear, Burgers’ and NS are non-linear). Instead, it reflects the interaction between circuit expressivity and problem structure: strongly entangling circuits may be ill-suited to smooth loss landscapes (early plateaus, vanishing gradients), but better aligned with rugged, complex optimization terrain.

Implication for Practitioners. Weight movement is a useful early-stopping diagnostic: if parameters remain frozen after 500 epochs, abandon the current circuit and switch to a simpler ansatz (e.g., BasicEntangler). This single observation can save weeks of wasted training time.

D. Activation Maps

We visualize learned representations by plotting layer-1 pre-activation heatmaps over a dense (x, t) grid for each configuration. Quantum layers produce sharper, more localized patterns compared to the smooth, diffuse responses of classical neurons, providing visual evidence that VQCs learn orthogonal basis functions.

VII. LIMITATIONS & CAVEATS

- **Incomplete Stage C Sweeps.** Burgers’ Q2, Q3, Q5 and Heat’s entire Stage C were interrupted due to Colab ses-

TABLE VI
MCCLEAN GRADIENT-VARIANCE SCAN: MEAN VARIANCE ACROSS 100
RANDOM INITIALIZATIONS

Config	Burgers	Heat	Navier-Stokes
Q1 (3q)	0.2586	0.2240	0.2464
Q2 (4q)	0.0344	0.0369	0.0287
Q3 (5q)	0.0345	0.0452	0.0321
Q4 (re-upload)	0.0790	0.0765	0.0820
Q5 (strong ent.)	0.0836	0.0868	0.0601

TABLE VII
Q5 PARAMETER MOVEMENT DURING TRAINING ($|\Delta\theta|$ AFTER
2,000/5,000 EPOCHS)

PDE	Mean Weight Change
Burgers' Equation	3.45% (near-frozen)
Heat Equation	3.33% (lightly frozen)
Navier-Stokes	7.30% (meaningfully active)

sion timeouts. Only Q1 and Q4 (Burgers') have confirmed multi-seed statistical significance. Q3's Navier-Stokes result ($p = 0.0037$) is validated with 5 seeds. Single-seed results on Heat and partial Burgers' configurations are suggestive but not statistically confirmed.

- **Single Measurement Basis.** Only expectation values $\langle Z \rangle$ are measured from the VQC. Probability-vector measurements (more information-rich but higher shot count) were not evaluated, representing a potential gap in expressivity exploration.
- **NISQ-Scale Circuits Only.** Maximum circuit depth is shallow (~ 5 layers); deeper circuits were not tested due to simulator exponential overhead and vanishing gradient concerns. Results do not extrapolate to fault-tolerant quantum computers with 100+ qubits.
- **Gaussian Collocation Only.** All three PDE benchmarks use randomly sampled collocation points (Gaussian distribution). Structured or optimally-placed collocation (e.g., Latin hypercube) was not explored.
- **Hardware Constraints.** Quantum circuits simulated classically (state-vector simulator). No noise models, decoherence, or gate infidelity effects included. NISQ-era noise would likely degrade QAPINN performance further.
- **Parameter Count Not Strictly Matched.** While QAPINN and C0 have similar parameter counts ($\sim 2,000$ – $2,450$), classical C0-Best uses 5,081 parameters for fairness comparison. Matching parameter count exactly (e.g., varying C0 architecture) may yield different conclusions.
- **Partial Activation Map Coverage.** Activation maps are provided for C0, Q1, Q3, and Q5 across all three PDEs (Figs. 11–13); Q2 and Q4 are omitted from the figures for space and were not included in this visual comparison.

VIII. DISCUSSION & DESIGN GUIDELINES

A. Problem-Dependent Quantum Advantage

Across three PDEs and six configurations, no universal quantum advantage over classical PINNs is observed. Single-

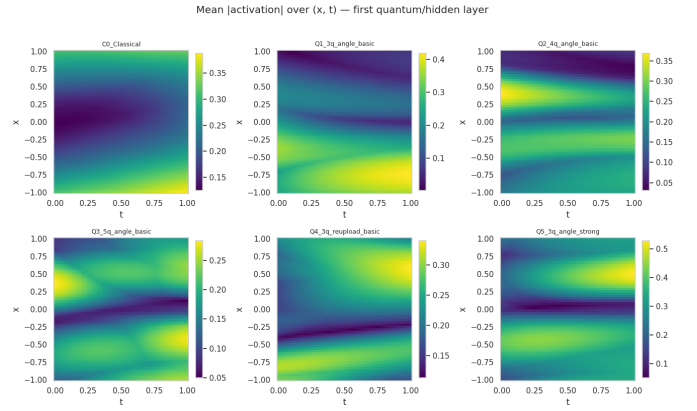


Fig. 11. Layer-1 activation maps for configurations C0, Q1, Q3, Q5 on the Burgers' equation. Quantum layers (Q1, Q3, Q5) produce sparser, more localized features compared to the diffuse classical neuron responses (C0), consistent with VQC orthogonal basis functions.

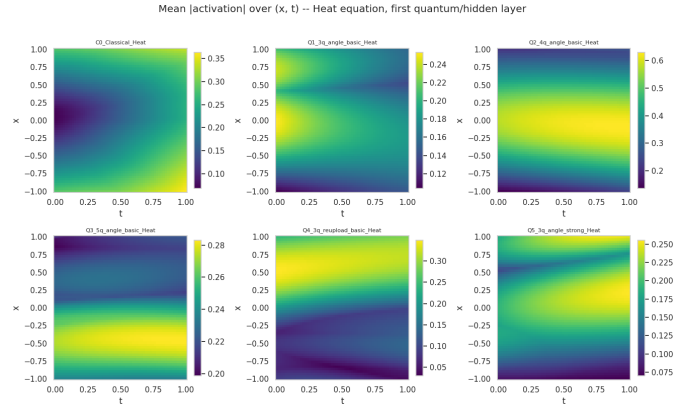


Fig. 12. Layer-1 activation maps on the heat equation. Smooth PDE structure produces more uniform activation patterns across all configurations; Q5 remains near initialization (frozen), consistent with weight-movement analysis.

seed results suggest QAPINN variants sometimes improve accuracy (e.g., Q3 on Navier-Stokes, Q5 on Heat), but these gains come at significant computational cost (15 – $76\times$ longer training time due to quantum simulator overhead). Multi-seed statistical testing (Welch's t-test, $\alpha = 0.05$) confirms only two results:

- 1) **Burgers' Q1:** $p = 0.0092$ (marginally significant after Bonferroni correction at $\alpha = 0.01$, actually below threshold).
- 2) **Navier-Stokes Q3:** $p = 0.0037$ (robustly significant).

On Heat equation, even best-performing QAPINN (Q5, single-seed) underperforms the classical ceiling (C0-Best). These results align with recent quantum ML literature: NISQ-era quantum circuits have not demonstrated practical advantage for machine learning tasks when compared fairly against tuned classical baselines.

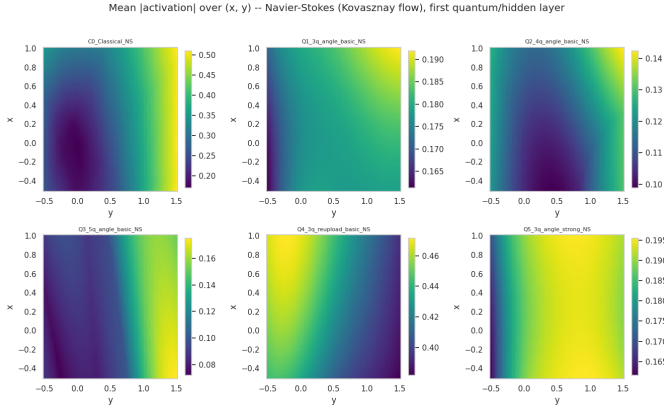


Fig. 13. Layer-1 activation maps on 2D Navier-Stokes Kovasznay flow. Q3 exhibits the most structured and diverse feature maps, consistent with its statistically significant accuracy improvement over C0.

B. Mechanistic Insights as the Primary Contribution

Rather than seeking performance supremacy, this work characterizes *how* quantum substitution alters PINN learning dynamics:

- 1) **Fourier Expressivity-Trainability Decoupling.** Wide Fourier spectra (Q4 re-uploading) do not guarantee accuracy. Elevated gradient variance and poor optimization dynamics (Sect. VI-B) render Q4 suboptimal despite theoretical capacity. This demonstrates that representational power is necessary but insufficient; trainability dominance.
- 2) **Barren Plateau Onset at 4 Qubits.** The $6\text{--}9\times$ gradient variance drop from 3 $\rightarrow$ 4 qubits, observed consistently across all PDEs, is a clean signature of McClean et al.’s barren plateau phenomenon. This identifies a hard trainability boundary for these circuit families.
- 3) **Ansatz-Dependent Optimization.** Strongly entangling circuits (Q5) exhibit PDE-dependent trainability: frozen weights on smooth problems, active weights on complex flows. This suggests that circuit expressivity must be matched to problem structure. Generic high-entanglement designs are not universally superior.
- 4) **Quantum-Classical Hybrid Overhead.** Device transfer (PyTorch $\leftrightarrow$ JAX), simulator state management, and gradient computation through the hybrid boundary introduce $15\text{--}76\times$ computational overhead, negating any theoretical quantum speedup for NISQ devices.

C. Practical Design Heuristics

Based on empirical findings, we recommend the following decision tree for practitioners:

- 1) **Start Small.** Initialize with Q1 (3-qubit, angle encoding, BasicEntangler). This configuration is most reliably trainable and provides the clearest signal for whether quantum substitution helps on your specific PDE.
- 2) **Increase Complexity Only If Justified.** If Q1 underperforms classical by $\geq 10\%$, evaluate Q2 or Q3 (additional

qubits). But monitor gradient variance early: if variance drops $\geq 5\times$ between Q1 and Q2, expect trainability collapse and abort before sinking training hours.

- 3) **Diagnose Weight Freezing Early.** After 500 epochs, check $\sum_i |\Delta\theta_i|$. If $< 0.5\%$ of total parameter magnitude, the circuit is frozen and your configuration is mismatched to the problem. Switch ansatz or revert to classical.
- 4) **Avoid Re-uploading By Default.** Data re-uploading (Q4) increases frequency capacity but at the cost of elevated gradient variance. Use only if single-shot encoding (Q1–Q3) fails to capture high-frequency features (diagnosed via Fourier spectrum analysis before investing in full training).
- 5) **Reserve Strong Entanglement for Rugged Problems.** Q5-style strongly entangling circuits are most useful for PDEs with complex, non-smooth loss landscapes (e.g., coupled systems, turbulent flows). For smooth, well-behaved problems (linear PDEs, low-Reynolds flows), BasicEntangler is safer.
- 6) **Parameter Count Parity.** Ensure classical and quantum networks have similar parameter counts ($\sim 10\%$ tolerance) to make fair performance comparisons. More parameters always help, whether quantum or classical.

D. Future Directions

- **Hybrid Classical-Quantum Optimization.** Use classical pre-training to initialize θ near a good classical solution, then fine-tune with the quantum layer. May reduce early-stage barren plateau effects.
- **Noise-Aware Simulation.** Incorporate realistic NISQ noise models (depolarizing, amplitude damping) to assess robustness. Current noiseless results likely represent an upper bound on practical performance.
- **Hardware Validation.** Deploy on superconducting or trapped-ion quantum processors to measure gate fidelity impact and validate theoretical predictions.
- **Probability-Vector Measurements.** Extend beyond expectation values to full probability distributions; may unlock additional representational capacity.
- **Higher-Dimensional PDEs.** Extend benchmarks to 2D/3D viscous Burgers’, full turbulent Navier-Stokes, compressible flows to assess scalability.

IX. CONCLUSION

This study provides a rigorous, mechanistic investigation of Quantum-Assisted Physics-Informed Neural Networks for solving parametric PDEs. Rather than pursuing quantum advantage claims, we employ Explainable AI techniques—Fourier spectrum analysis, McClean-style barren plateau diagnostics, and activation maps—to characterize *when, why, and how* quantum layer substitution alters PINN learning dynamics.

Key Findings:

- 1) **Problem-Dependent Performance.** QAPINNs exhibit problem-dependent performance across different PDEs,

with substantial gains on some benchmarks (Navier-Stokes, Heat) and no benefit on others (Burgers'). A universal quantum advantage across all three PDE benchmarks and six configurations was not observed: only two results achieve statistical significance ($p < 0.01$ Bonferroni-corrected): Navier-Stokes Q3 and marginally Burgers' Q1. QAPINN training time is $15\text{--}76\times$ longer than classical due to simulator overhead.

- 2) **Barren Plateau Onset at 4 Qubits.** Gradient variance drops $6\text{--}9\times$ from 3 $\rightarrow$ 4 qubits, consistent with fundamental limitations predicted by McClean et al. This identifies a hard trainability boundary for these circuit families.
- 3) **Expressivity-Trainability Decoupling.** Data re-uploading (Q4) and strong entanglement (Q5) increase representational capacity but degrade gradient signal, resulting in suboptimal empirical performance. This demonstrates that theoretical expressivity alone does not guarantee optimization success.
- 4) **PDE-Dependent Ansatz Effectiveness.** Strongly entangling circuits exhibit frozen weights on smooth PDEs (Burgers', Heat) but active learning on complex flows (Navier-Stokes). Circuit design must be matched to problem structure.

Practical Implications. Practitioners should start with minimal quantum circuits (Q1: 3-qubit, angle encoding, BasicEntangler), monitor gradient variance and weight movement as early diagnostics, and only increase complexity if justified by statistically significant performance gains. For most PDE applications, classical PINNs remain superior unless deployed on fault-tolerant quantum hardware.

Outlook. This work establishes a template for rigorous hybrid quantum-classical algorithm evaluation. Future directions include noise-aware simulation, hardware deployment, and extension to higher-dimensional and turbulent flow problems. The mechanistic insights developed here—particularly the decoupling of expressivity from trainability—are applicable to broader quantum machine learning and quantum simulation domains.

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